

Actualizacion de Resultados: BL nuevo vs sin BL

Generado el 27-06-2026.

Configuracion de la corrida

- BL nuevo = robust_factor_hybrid: 65% factor BL y 35% asset momentum.
- Sin BL = optimizacion con retornos historicos, sin ajuste Black-Litterman.
- Lambda = 0, train window = 5 anos, full_invest = True.
- max_weight comparados = 1%, 2,5%, 5%, 10%, 15% y 20%.
- Retornos anuales independientes: 2019 a 2025. MAE: 2019 a 2024.
- La columna 2,5% aparece destacada porque fue la restriccion recomendada.

Lectura del delta: positivo favorece al BL en retorno y aceptacion; negativo favorece al BL en MAE, abandono y drawdown.

Retorno Promedio

Retorno promedio por perfil

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
Conservador	-0.7%	-0.7%	-0.7%	-0.7%	-0.7%	-0.7%
Neutro	4.7%	4.0%	3.5%	3.5%	3.5%	3.5%
Arriesgado	17.1%	19.9%	19.2%	19.4%	19.1%	19.0%
Muy arriesgado	20.1%	23.7%	24.9%	20.7%	20.9%	20.8%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
Conservador	-0.7%	-0.7%	-0.7%	-0.7%	-0.7%	-0.7%
Neutro	4.9%	4.7%	4.9%	4.9%	4.9%	4.9%
Arriesgado	18.2%	17.6%	17.3%	17.5%	17.9%	18.0%
Muy arriesgado	21.5%	23.6%	21.7%	22.8%	24.1%	24.6%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	-0.0%	-0.0%	-0.0%	-0.0%	-0.0%	-0.0%
Neutro	-0.2%	-0.7%	-1.4%	-1.3%	-1.3%	-1.4%
Arriesgado	-1.0%	2.3%	1.9%	1.9%	1.2%	1.0%
Muy arriesgado	-1.4%	0.1%	3.2%	-2.1%	-3.2%	-3.8%

Error Predictivo

MAE por perfil

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	4.3%	4.3%	4.3%	4.3%	4.3%	4.3%
Neutro	4.2%	4.0%	4.0%	4.0%	4.0%	4.0%
Arriesgado	3.4%	3.7%	3.9%	3.8%	3.9%	3.9%
Muy arriesgado	4.1%	4.4%	4.9%	4.8%	4.8%	4.8%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	4.2%	4.2%	4.2%	4.2%	4.2%	4.2%
Neutro	4.1%	4.0%	3.9%	3.9%	3.9%	3.9%
Arriesgado	3.7%	4.0%	4.0%	3.9%	3.9%	3.9%
Muy arriesgado	4.4%	5.1%	5.0%	4.8%	4.8%	4.8%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
Neutro	0.1%	0.0%	0.1%	0.1%	0.1%	0.1%
Arriesgado	-0.3%	-0.3%	-0.1%	-0.1%	0.0%	0.0%
Muy arriesgado	-0.3%	-0.7%	-0.1%	-0.0%	-0.0%	0.0%

Probabilidades

Probabilidad de aceptacion

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
Conservador	89.5%	89.5%	89.5%	89.5%	89.5%	89.5%
Neutro	56.7%	68.0%	67.4%	67.5%	67.5%	67.5%
Arriesgado	28.3%	48.3%	55.4%	58.0%	58.5%	58.6%
Muy arriesgado	26.8%	47.2%	58.6%	62.2%	63.0%	63.2%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
Conservador	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Neutro	84.0%	99.2%	99.1%	99.1%	99.1%	99.1%
Arriesgado	53.7%	94.9%	99.5%	99.8%	99.9%	99.9%
Muy arriesgado	30.6%	94.9%	99.9%	100.0%	100.0%	100.0%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	-10.5%	-10.5%	-10.5%	-10.5%	-10.5%	-10.5%
Neutro	-27.2%	-31.2%	-31.7%	-31.6%	-31.6%	-31.6%
Arriesgado	-25.3%	-46.5%	-44.1%	-41.8%	-41.4%	-41.3%
Muy arriesgado	-3.8%	-47.7%	-41.3%	-37.7%	-37.0%	-36.8%

Probabilidad de abandono

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	24.6%	24.6%	24.6%	24.6%	24.6%	24.6%
Neutro	7.3%	7.3%	7.3%	7.3%	7.3%	7.3%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	24.4%	24.4%	24.4%	24.4%	24.4%	24.4%
Neutro	7.3%	7.3%	7.3%	7.3%	7.3%	7.3%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Neutro	-0.0%	-0.0%	-0.0%	-0.0%	-0.0%	-0.0%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Tasa de abandono observada

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	71.4%	71.4%	71.4%	71.4%	71.4%	71.4%
Neutro	28.6%	28.6%	28.6%	28.6%	28.6%	28.6%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	71.4%	71.4%	71.4%	71.4%	71.4%	71.4%
Neutro	28.6%	28.6%	28.6%	28.6%	28.6%	28.6%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Neutro	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Muy arriesgado	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Drawdown Promedio

Max drawdown promedio

Delta = Con BL nuevo - Sin BL.

Con BL nuevo

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%
Neutro	9.9%	9.7%	9.9%	9.9%	9.9%	9.9%
Arriesgado	9.0%	10.1%	11.0%	11.5%	11.7%	11.7%
Muy arriesgado	11.8%	12.2%	12.8%	13.9%	14.3%	14.5%

Sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	6.3%	6.3%	6.3%	6.3%	6.3%	6.3%
Neutro	9.9%	9.5%	9.4%	9.4%	9.4%	9.4%
Arriesgado	10.1%	11.2%	11.4%	11.3%	11.1%	11.1%
Muy arriesgado	11.8%	13.6%	14.2%	13.7%	13.6%	13.5%

Delta BL - sin BL

Perfil	1%	2,5%	5%	10%	15%	20%
Muy conservador	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Conservador	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
Neutro	0.0%	0.2%	0.5%	0.5%	0.5%	0.5%
Arriesgado	-1.1%	-1.2%	-0.4%	0.2%	0.5%	0.6%
Muy arriesgado	0.0%	-1.4%	-1.4%	0.2%	0.7%	1.0%

Retornos por Ano - Con BL Nuevo

Cada ano parte nuevamente desde capital inicial.

Con BL nuevo - Muy conservador

Muy conservador	1%	2,5%	5%	10%	15%	20%
2019	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2020	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2021	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2022	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2023	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2024	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2025	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%

Con BL nuevo - Conservador

Conservador	1%	2,5%	5%	10%	15%	20%
2019	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%
2020	-33.5%	-33.5%	-33.5%	-33.5%	-33.5%	-33.5%
2021	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2022	-14.7%	-14.7%	-14.7%	-14.7%	-14.7%	-14.7%
2023	-10.2%	-10.2%	-10.2%	-10.2%	-10.2%	-10.2%
2024	21.4%	21.4%	21.4%	21.4%	21.4%	21.4%
2025	21.8%	21.8%	21.8%	21.8%	21.8%	21.8%

Con BL nuevo - Neutro

Neutro	1%	2,5%	5%	10%	15%	20%
2019	21.1%	15.6%	13.9%	13.7%	13.7%	13.6%
2020	-65.1%	-64.8%	-66.5%	-66.3%	-66.3%	-66.4%
2021	39.3%	39.3%	39.3%	39.3%	39.3%	39.3%
2022	-21.3%	-21.3%	-21.3%	-21.3%	-21.3%	-21.3%
2023	14.9%	14.9%	14.9%	14.9%	14.9%	14.9%
2024	21.4%	21.4%	21.4%	21.4%	21.4%	21.4%
2025	22.7%	22.7%	23.0%	23.0%	23.0%	23.0%

Con BL nuevo - Arriesgado

Arriesgado	1%	2,5%	5%	10%	15%	20%
2019	25.6%	18.8%	14.3%	9.2%	9.7%	9.0%
2020	29.9%	38.8%	47.6%	54.0%	51.0%	51.1%
2021	35.8%	25.4%	22.1%	21.8%	22.0%	21.9%
2022	-11.4%	-10.3%	-7.9%	-8.8%	-10.0%	-10.0%
2023	5.7%	13.4%	17.2%	19.4%	19.5%	19.4%
2024	21.5%	27.5%	27.0%	28.8%	28.6%	28.6%
2025	12.8%	25.6%	13.9%	11.6%	12.7%	12.8%

Con BL nuevo - Muy arriesgado

Muy arriesgado	1%	2,5%	5%	10%	15%	20%
2019	31.5%	25.7%	18.1%	11.9%	8.8%	7.7%
2020	37.0%	53.1%	63.6%	53.0%	55.8%	54.3%
2021	39.1%	29.9%	23.2%	16.6%	16.3%	16.6%
2022	-17.7%	-13.4%	-12.2%	-13.5%	-14.6%	-15.1%
2023	15.8%	20.9%	23.7%	21.7%	22.3%	23.6%
2024	24.6%	27.7%	29.4%	32.0%	33.5%	32.6%
2025	10.5%	22.2%	28.6%	23.3%	24.5%	25.9%

Retornos por Ano - Sin BL

Misma configuracion, pero sin aplicar Black-Litterman.

Sin BL - Muy conservador

Muy conservador	1%	2,5%	5%	10%	15%	20%
2019	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2020	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2021	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2022	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2023	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2024	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%
2025	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%	-0.9%

Sin BL - Conservador

Conservador	1%	2,5%	5%	10%	15%	20%
2019	10.3%	10.3%	10.3%	10.3%	10.3%	10.3%
2020	-33.5%	-33.5%	-33.5%	-33.5%	-33.5%	-33.5%
2021	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2022	-14.7%	-14.7%	-14.7%	-14.7%	-14.7%	-14.7%
2023	-10.2%	-10.2%	-10.2%	-10.2%	-10.2%	-10.2%
2024	21.4%	21.4%	21.4%	21.4%	21.4%	21.4%
2025	21.8%	21.8%	21.8%	21.8%	21.8%	21.8%

Sin BL - Neutro

Neutro	1%	2,5%	5%	10%	15%	20%
2019	22.4%	19.1%	18.1%	17.8%	17.8%	17.8%
2020	-64.5%	-62.7%	-60.4%	-60.2%	-60.2%	-60.2%
2021	39.3%	39.3%	39.3%	39.3%	39.3%	39.3%
2022	-21.7%	-21.7%	-21.7%	-21.7%	-21.7%	-21.7%
2023	15.5%	15.5%	15.5%	15.5%	15.5%	15.5%
2024	21.4%	21.4%	21.4%	21.4%	21.4%	21.4%
2025	21.8%	21.8%	22.1%	22.0%	22.0%	22.0%

Sin BL - Arriesgado

Arriesgado	1%	2,5%	5%	10%	15%	20%
2019	29.9%	32.7%	27.5%	29.8%	30.1%	29.9%
2020	29.9%	38.8%	38.9%	38.3%	38.4%	38.0%
2021	39.6%	33.4%	35.7%	35.4%	35.5%	35.3%
2022	-15.5%	-22.1%	-23.3%	-22.5%	-22.2%	-22.2%
2023	8.0%	10.1%	9.0%	8.5%	10.2%	11.2%
2024	18.6%	21.8%	25.4%	24.5%	25.0%	25.4%
2025	16.7%	8.4%	7.8%	8.5%	8.3%	8.3%

Sin BL - Muy arriesgado

Muy arriesgado	1%	2,5%	5%	10%	15%	20%
2019	32.8%	39.6%	36.1%	38.4%	40.6%	40.7%
2020	35.1%	60.2%	61.6%	56.3%	56.6%	57.0%
2021	39.4%	34.2%	37.8%	38.2%	38.2%	38.3%
2022	-20.2%	-28.9%	-33.9%	-31.3%	-30.6%	-30.2%
2023	20.9%	19.1%	15.4%	14.4%	16.2%	19.3%
2024	20.3%	21.9%	25.3%	29.3%	31.6%	32.1%
2025	22.1%	18.9%	9.8%	14.2%	16.3%	14.8%

Diferencia Anual: Con BL - Sin BL

Delta positivo significa que el BL nuevo tuvo mayor retorno que el modelo sin BL en ese año.

Delta anual - Muy conservador

Muy conservador	1%	2,5%	5%	10%	15%	20%
2019	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2020	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2021	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2022	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2023	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2024	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2025	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Delta anual - Conservador

Conservador	1%	2,5%	5%	10%	15%	20%
2019	-0.3%	-0.3%	-0.3%	-0.3%	-0.3%	-0.3%
2020	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2021	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2022	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2023	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2024	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2025	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Delta anual - Neutro

Neutro	1%	2,5%	5%	10%	15%	20%
2019	-1.3%	-3.6%	-4.2%	-4.1%	-4.1%	-4.2%
2020	-0.6%	-2.1%	-6.1%	-6.1%	-6.1%	-6.1%
2021	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2022	0.4%	0.4%	0.4%	0.4%	0.4%	0.4%
2023	-0.6%	-0.6%	-0.6%	-0.6%	-0.6%	-0.6%
2024	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2025	0.9%	0.9%	0.9%	1.0%	1.0%	1.0%

Delta anual - Arriesgado

Arriesgado	1%	2,5%	5%	10%	15%	20%
2019	-4.3%	-13.9%	-13.2%	-20.6%	-20.5%	-21.0%
2020	0.0%	-0.0%	8.6%	15.7%	12.6%	13.1%
2021	-3.7%	-8.1%	-13.5%	-13.6%	-13.5%	-13.4%
2022	4.1%	11.8%	15.4%	13.6%	12.2%	12.2%
2023	-2.3%	3.3%	8.2%	10.9%	9.3%	8.2%
2024	2.9%	5.7%	1.6%	4.3%	3.6%	3.1%
2025	-3.9%	17.2%	6.1%	3.1%	4.5%	4.5%

Delta anual - Muy arriesgado

Muy arriesgado	1%	2,5%	5%	10%	15%	20%
2019	-1.3%	-13.9%	-18.1%	-26.4%	-31.8%	-33.0%
2020	1.9%	-7.2%	2.0%	-3.3%	-0.8%	-2.7%
2021	-0.3%	-4.2%	-14.6%	-21.6%	-21.9%	-21.8%
2022	2.5%	15.5%	21.7%	17.8%	16.0%	15.2%
2023	-5.1%	1.8%	8.3%	7.3%	6.1%	4.4%
2024	4.2%	5.7%	4.1%	2.7%	1.9%	0.5%
2025	-11.6%	3.3%	18.8%	9.1%	8.2%	11.0%

Lectura Para el Informe

- El 2,5% sigue siendo la restriccion elegida por equilibrio: diversifica, evita concentracion extrema y mantiene retornos competitivos.
- El BL nuevo ayuda especialmente al perfil arriesgado en 2,5%, 5%, 10%, 15% y 20%, pero no supera a sin BL en neutro.
- En muy arriesgado, el BL nuevo mejora en 5%, pero pierde contra sin BL en los pesos mas altos.
- En conservador y muy conservador las diferencias de retorno son casi cero porque las restricciones/probabilidades dominan mas que el ajuste BL.
- El MAE debe leerse como error promedio entre prediccion y retorno real mensual; menor MAE es mejor.

Archivos fuente

Archivo	Contenido usado
max_weight_hybrid_vs_sin_bl_metricas_anuales.csv	Retornos por ano, probabilidades y abandonos por perfil/max_weight/modelo.
max_weight_hybrid_vs_sin_bl_error_predictivo.csv	MAE, RMSE, bias e hit rate por perfil/max_weight/modelo.
max_weight_hybrid_vs_sin_bl_resumen.csv	Metricas promedio y deltas con BL nuevo menos sin BL.